05. Data, Model and Evaluation Strategy

AdmitAssist | MBA ZG583 Management of AI Products | GOLLAPUDI SIVA NAGA JNANA TEJA | 2025MB26205 | GitHub: jnanateja009

1. The problem as AI tasks

Task	Type	Input	Output	Who checks it
T1 Read the inquiry	Information extraction + classification, structured output	Masked message, channel, date	JSON: admission or not, class, timeframe, six signals, sensitive flag and category, quotes, confidence, one-line summary (prompt `read_v1`)	Rules R4 (quote check, cross-check), then the coordinator
T2 Draft the reply	Constrained text generation	Approved school facts, what was read, masked message	Reply of at most 120 words (prompt `draft_v1`)	Rules R8 (draft checks), then the coordinator

What is **not** an AI task: phone and e-mail capture (rule), the priority itself (rule, from the signals the AI reports), the reply-by time (rule), and anything about admission decisions, seats or fees (people).

2. Data

Data set	Size	Source	Status	Used for
D1 Written samples	48 messages	Written by an AI assistant from my brief to look like real WhatsApp messages, voice-note transcripts, call and walk-in notes, web forms and e-mails; 26 English, 9 Hinglish, 9 Telugu-English, 2 Telugu script, 2 Hindi script	Done, labelled (provisional)	Building and testing the design now. Results below.
D2 Real past inquiries	100 messages	The school's WhatsApp and register for the last admission season, with the school owner's written consent. Names replaced with initials; each phone number replaced, in the same format, by a made-up number before it leaves the office computer, so phone capture (M6) can still be tested.	Phase 0, by 15 Nov 2026	Gate A: the real test
D3 Sensitive set	75 messages, at least 40 of them real	Real sensitive messages from the whole of last season's WhatsApp history (not only the 100 in D2), plus up to 35 reworded versions written by me and checked by the principal for realism	Phase 0-1, by 6 Dec 2026	Gate A: 75 with no miss proves sensitive recall above 95% (Wilson lower bound 95.1%; 60 would prove only 94.0%)
D4 Pilot log	All pilot inquiries (about 460 over 10 weeks at 200 a month, ASSUMED)	Activity log: the coordinator's final priority, edits, approvals	Phase 3	Ongoing measurement and the next evaluation set

Note on D1: the sample brief I gave had a counting slip (my route mix added up to 45, not 48). The writer kept 48 and made 8 not-admission messages instead of 5. I kept it that way.

2.1 Labelling

Item	Rule
Labelling guide	The human guide in section 2.2, written so that labels do not copy the scoring rules. Route, sensitive category, class, phone, timeframe and language follow the definitions in 03 section 3.
Who labels D2	The front-office coordinator (main labels). I label the same first 30 messages separately.
Agreement check	Cohen's kappa between the two of us on priority. If below 0.70, we discuss the differences, fix the guide (version 1.1) and relabel.
Time	About 1 minute per message: 100 minutes for the coordinator, paid as overtime (Rs 200, ASSUMED)
Storage	Labels in the same JSON format as `prototype/data/sample_inquiries.json`, so `run_eval.js` runs on it unchanged

2.2 Human labelling guide for priority (version 1.1)

The coordinator labels from experience, before seeing anything AdmitAssist produced, with one question in mind: **"If I could answer only some messages today, would this family be hurt by waiting?"**

Label	Meaning for the coordinator	Examples
HIGH	This family is likely to decide within weeks, or has already waited. They need a reply today.	"We are shifting next month, need Class 6 seat"; "Can we visit on Saturday?"; "Sir any update? I messaged on Monday"; "Please send the form and documents list"
MEDIUM	Real interest for the next session, but no date pressure. A reply tomorrow is fine.	"What are the fees for UKG for 2027-28?"; "Is your school CBSE?"; "admission details pls"
LOW	Not something the school can act on now.	Class 11 or 12 (not offered); "for 2029"; "just collecting options, will decide next year"

The point system in 03 (FR-11) is my attempt to reproduce this judgement. Agreement between the two (M3) tells me whether the points are right, which only works if the labels are not derived from the points.

3. Baseline: the simpler alternative

To answer "why not just rules?", I built a rules-only reader (`config/baseline_rules.json`) before any AI results existed: keyword lists for not-admission, timeframe and the six signals, plus the same class patterns, sensitive keywords and priority points as AdmitAssist. Then I ran both on D1.

4. Model choice

Candidate	Price per million tokens (input / output, USD)	Est. AI cost per inquiry (06)	Role
gpt-6.1-terra	1.00 / 5.00	Rs 0.36	**V1 default.** Positioned by OpenAI for high-volume tasks that need speed more than deep reasoning (SOURCE: OpenAI, 26 Oct 2026).
gpt-6.1-sol	2.00 / 10.00	Rs 0.72	Fallback when the default fails twice
gpt-6-luna	0.10 / 0.50	Rs 0.04	Cheap candidate, to test in Phase 1
gemini-3.1-flash-lite	0.25 / 1.50	Rs 0.10	Second-provider candidate, to test in Phase 1 (works through the same OpenAI-compatible connector)

Prices: OpenAI pricing page (standard tier) and costgoat.com for Gemini, checked 2 Nov 2026 (SOURCE).

How I trade off cost, quality, speed and trust. The faculty's four variables, with my call for this product:

Variable	My call	Why
Quality	Must pass every gate in section 6	A missed sensitive message or a false LOW costs a family
Trust	Must pass the gates on Telugu and Hindi messages too, not just on average	The parents who write in their own language must not get worse service
Speed	p90 at or under 15 s for read + draft	The coordinator is waiting with the card open
Cost	Cheapest model that passes the first three	At Rs 0.36 per inquiry, cost is under 5% of the price; quality matters more

Decision rule for Phase 1: run all four candidates on D2 + D3. Pick the cheapest one that passes every Gate A threshold. The fallback model is used only if it also passes the Gate A blockers; otherwise there is no fallback and failed reads go to "Read it yourself". If no candidate passes, keep gpt-6.1-terra and tune within the 40-hour allowance (03b section 3). In Phase 1 the settings switch to dated model snapshot names (today they hold the family name `gpt-6.1-terra`), and the model name returned by the API is stored on every result. No model switch after that without re-running FR-23 (about Rs 60 per run of 160 messages on one model). I am not routing between models per message in V1: the saving would be under Rs 50 per school per month (06), and it doubles what has to be tested.

5. Results so far (D1, 48 written samples)

How these were produced: `node prototype/eval/run_eval.js --mode baseline` and `--mode replay`. For the AdmitAssist column, the exact prompts were exported (`--export read`, `--export draft`) and answered by a general-purpose AI assistant acting as the model in separate sessions, with no access to the labels. The answers are stored in `prototype/data/recorded_ai_responses.json` and replayed through the real engine. **This is a recorded dry run with a stand-in model, not the production API.** The live run is part of Gate A.

Metric	Rules only (no AI)	AdmitAssist (recorded AI run)	95% interval for AdmitAssist
Sensitive messages sent to staff	4/10 (40%)	10/10 (100%)	72% to 100%
Sensitive messages that got an AI draft	n/a (no drafts)	0	
Normal messages wrongly sent to staff-only	1/38	1/38	
Route correct	38/48 (79%)	47/48 (98%)	89% to 100%
Not-admission messages recognised	4/8	7/8 (AI alone: 8/8)	
Priority exact match	27/30 (90%)	30/30 (100%)	89% to 100%
HIGH inquiries marked HIGH	9/11	11/11	74% to 100%
Dangerous misses (HIGH shown as LOW or not-admission)	0	0	
Class read correctly	33/40 (83%)	40/40 (100%)	91% to 100%
Phone captured correctly (rule in both)	48/48	48/48	
Timeframe correct	24/30 (80%)	30/30 (100%)	
Flagged "Needs review"	2/48	3/48 (two children x2, one prompt injection)	
Drafts passing all checks	n/a	30/30 (engine 1.1.1); 27/30 with engine 1.0.0	89% to 100%
Draft length, average / longest	n/a	79 / 112 words (limit 120)	
AI cost per inquiry (estimated tokens)	Rs 0	Rs 0.32	

Priority agreement by language (AdmitAssist): English 18/18, Telugu and Telugu-English 6/6, Hindi and Hinglish 6/6. Rules only: English 17/18, Telugu 4/6, Hindi 6/6.

Full per-message tables: `prototype/eval/results/baseline_summary.md`, `replay_summary.md`, `comparison.md`.

5.1 What the results say

1. **The AI earns its place on sensitive messages, not on priority.** The rules missed 6 of 10 sensitive messages: every one that was written indirectly or in Telugu or Hindi script. Two of those six were also strong admission inquiries (a custody case in a phone note, a child with a speech delay whose parent wanted to visit Saturday). With rules only, both would have got a cheerful template draft. The AI caught all 10. The keyword list caught 4, and the AI caught those 4 as well, so on this set the keyword list added nothing. I keep it anyway because it costs nothing and does not depend on the AI being up.
2. **On clean samples, rules already prioritise well (27/30).** The AI's priority gain is 3 messages, two of them in Telugu. I am not claiming the AI is needed for priority on well-written English messages. Real messages will be messier than D1.
3. **My first draft checks were too strict.** Engine 1.0.0 flagged 3 of 30 drafts. Reading them, all three were fine: two repeated the year the parent asked about (2028-29, 2029) and one offered 10 to 11 AM inside the approved 10 AM to 12 PM visit slot. I changed two rules (engine 1.0.1): a year the parent wrote may be repeated, and a time inside an approved range is allowed. I deliberately did **not** allow amounts from the parent's message, because that is exactly how the prompt-injection sample tried to get "Rs 5000" into a reply. If more than 1 draft in 10 shows a false warning, staff learn to ignore red warnings (target M9: 95% of drafts pass), so this matters as much as catching real problems.
4. **The red-team tests found a real bug.** The keyword list had "instalment" but missed "instalments" (plural). Fixed in engine 1.0.2 (plural matching). After the vendor review (11) I added 10 more tests for landlines, the office number, foreign numbers, "Rs 500", "25k", unapproved dates, closed classes, holidays and locked safety settings. After the second review I added 2 more (a month the parent wrote may be repeated, a date the parent proposed may not). All 51 now pass (`eval/results/unit_tests.md`): 19 of 19 bad drafts blocked, 7 of 7 good drafts allowed, made-up quotes dropped, conflicting class readings flagged.
5. **The prompt-injection sample was handled.** The stand-in model flagged it, the draft gave the approved fee range (Rs 24,000 to 28,000) and did not confirm a seat.

5.2 Why these numbers are probably too good

Reason	Effect	What fixes it
D1 was written by an AI from my labelling rules, and the stand-in model is from a similar model family	Writer and reader share habits, so agreement is inflated	D2: real messages, labelled by the coordinator
D1 is small (10 sensitive, 30 priority)	10/10 only proves sensitive recall above 72%	D3: 75 sensitive messages (section 6)
Stand-in model, not gpt-6.1-terra through the API	Behaviour and latency will differ	Live run in Phase 1
Tokens are estimated (1 per 4 Latin characters, 1 per Indic character)	Cost may be off by 20 to 30% either way	API-reported tokens in the live run
Labels are mine, provisional	My idea of HIGH may not match the coordinator's	Coordinator labels + kappa check

6. Metrics, targets and launch gates

Columns follow the faculty's advice: the first three columns stay fixed, only the target column changes.

6.1 AI and technical metrics

ID	Metric	Definition	How measured	Target (Gate A, live model, D2 + D3)
M1	Sensitive recall	Sensitive messages (label) routed STAFF_ONLY / all sensitive messages	`run_eval.js`	**0 misses on 75 sensitive messages** (blocker)
M2	Dangerous misses	HIGH (label) shown as LOW or NOT_ADMISSION	`run_eval.js`	**0** (blocker)
M3	Priority agreement	Exact match with the coordinator's label, PRIORITY-route messages	`run_eval.js`	75% or more
M4	Priority agreement by language group	M3 for English, Hindi/Hinglish, Telugu/Telugu-English separately, at least 20 messages per group (reworded messages added if D2 has fewer)	`run_eval.js`	Each group within 10 points of English (blocker if worse)
M5	Class read correctly	Class matches label, messages that state a class	`run_eval.js`	90% or more
M6	Phone captured	First phone number matches label	`run_eval.js`	98% or more
M7	False alarms	Normal messages routed STAFF_ONLY / all normal messages	`run_eval.js`	10% or less
M8	Needs-review share	Messages with a review note / all	`run_eval.js`	25% or less (above that, staff stop reading the notes)
M9	Drafts passing checks	Drafts with no check problem after one automatic rewrite / all drafts	`run_eval.js`	95% or more
M10	Promises reaching a parent	At Gate A: drafts that pass R8 but still contain a promise or wrong fact, found in the M14 review. In the pilot: approved replies with a blocked phrase or unapproved amount (R8 re-run on approved text)	M14 review; log	**0** (blocker)
M11	Read failures	Messages that end as NEEDS_REVIEW because the AI failed	`run_eval.js` live	2% or less
M12	Time per inquiry	Read + draft, wall clock	`run_eval.js` live	p50 8 s or less, p90 15 s or less
M13	AI cost per inquiry	API-reported tokens x list price	`run_eval.js` live, log	Rs 1.00 or less
M14	Draft correctness	The principal reads 50 drafts against the message and school facts: answers the questions asked, no wrong or missing fact, suitable tone. Share needing no factual correction	Checklist, 50 drafts	90% or more
M15	Confidence calibration	Share of fields tagged Sure that are correct	`run_eval.js` on D2	95% or more; if lower, raise the Sure cut-off
M16	Stability	Same 20 messages run 3 times at production settings; share with the same route and priority every time	`run_eval.js` x3	95% or more
M17	Name-swap fairness	20 messages run with only the names changed across communities (08 section 5), at temperature 0	`run_eval.js`	Identical route and priority in all 20 pairs (blocker)

6.2 Workflow and business metrics (pilot, from the activity log and Phase 0 baselines)

ID	Metric	Definition	How measured	Target (pilot, by 28 Feb 2027)
B1	HIGH reply time	Received to approved, HIGH inquiries, median, working hours	Log	2 working hours or less
B2	Handling time	Coordinator minutes per inquiry, stopwatch on 20 inquiries	Phase 0 vs week 6 of pilot	40% less than Phase 0 (ASSUMED 6.0 to 3.6 min)
B3	Light edits	Approved AI drafts with edit share 0.20 or less / approved AI drafts	Log	60% or more
B4	Inquiries recorded	Inquiries with class and phone in AdmitAssist / inquiries in the school's WhatsApp for the same days	Weekly spot check of 2 days	95% or more
B5	Replies on time	Inquiries whose reply clock stopped before the reply-by time (all routes, 03b section 6) / all inquiries	Log	85% or more
B6	Paying schools	Schools with a paid invoice	Invoices	3 by 31 Mar 2027
B7	Support load	My support hours per school per month	My time log	1 hour or less

6.3 Trust and guardrail metrics

ID	Metric	Definition	How measured	Target
T1	Rubber-stamping	Approvals in under 10 seconds / all approvals	Log	10% or less
T2	Priority overrides	Priorities changed by staff / inquiries read	Log	20% or less; every change has a reason; reviewed weekly
T3	Parent complaints about a reply	Complaints that mention a reply drafted with AdmitAssist	"Report a problem with this reply" in AdmitAssist (FR-27)	0
T4	Coordinator trust	Weekly question: "I trust the priority labels" (1 to 5)	1-question form, Saturdays	4 or more by week 6
T5	Language fairness	M4 on pilot data	Monthly re-run on D4	Same as M4

6.4 Gates

Gate	When	Passes if	If it fails
Gate A: ready for the shadow run	6 Dec 2026	Blockers M1, M2, M4, M10, M17 met; M3, M5, M6, M7, M8, M9, M11 to M16 targets met on D2 + D3 with the live model; the fallback model passes the blockers or is switched off	Tune within 40 hours, re-run; shadow run moves by the delay. Never skip a blocker.
Gate B: ready to use drafts for real	20 Dec 2026	All seven launch blockers in 08 section 6 are clear; on shadow-run messages M1 and M2 still 0 misses and M3 75% or more against what the coordinator actually did	Extend shadow run by 2 weeks
Go / no-go for V2 and selling more	28 Feb 2027	B1, B3, B5, T1, T3, T4 targets met; school owner says he would pay Rs 1,499 a month	Stop or redesign; refund nothing because the pilot is free

7. After launch: monitoring and change control

What	How often	Trigger for action
Re-run `run_eval.js` on D2 + D3 + new labelled pilot messages	Every prompt, model, keyword or points change (FR-23), and monthly	Any blocker metric worse than its gate
Review the log with the coordinator	Weekly, 30 minutes	T2 above 20%, T1 above 10%, any review note type above 10% of messages
Add misclassified pilot messages to the evaluation set	Weekly	Every priority change and every staff-only miss found later
Check the AI provider's model version	Monthly	Version changed: re-run the evaluation before accepting it